

Teaching Exam

Subject – Computer Science

**Web Based Application
Development**



Lecture No –1

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Topics to be Covered



Topic

Web based Application

Browser

①

Canva.
Google.doc

CRED

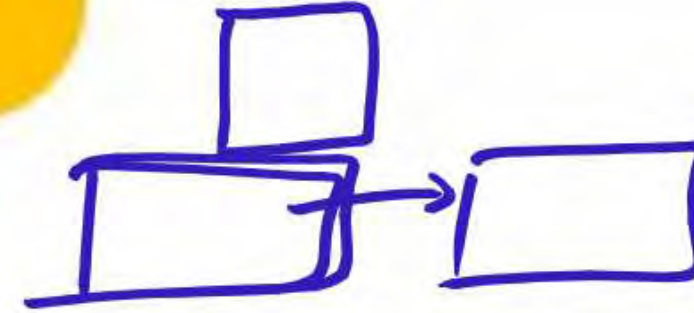
②

PUBG

③

X

Figma



Mobile Application.

System Software:

↓
Operating System

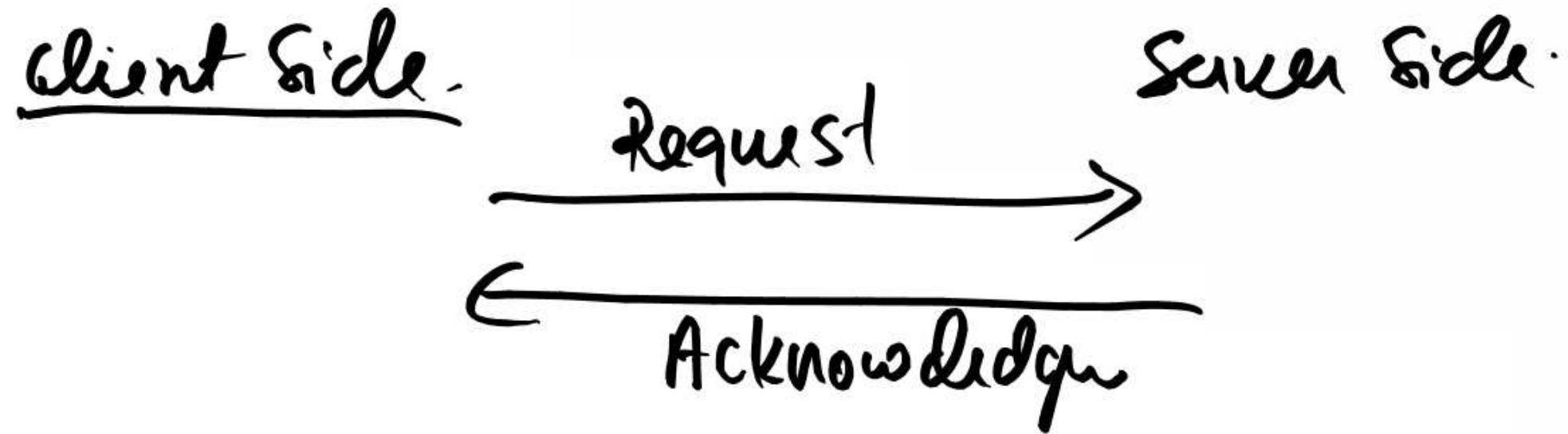
Drivers

web based applications.

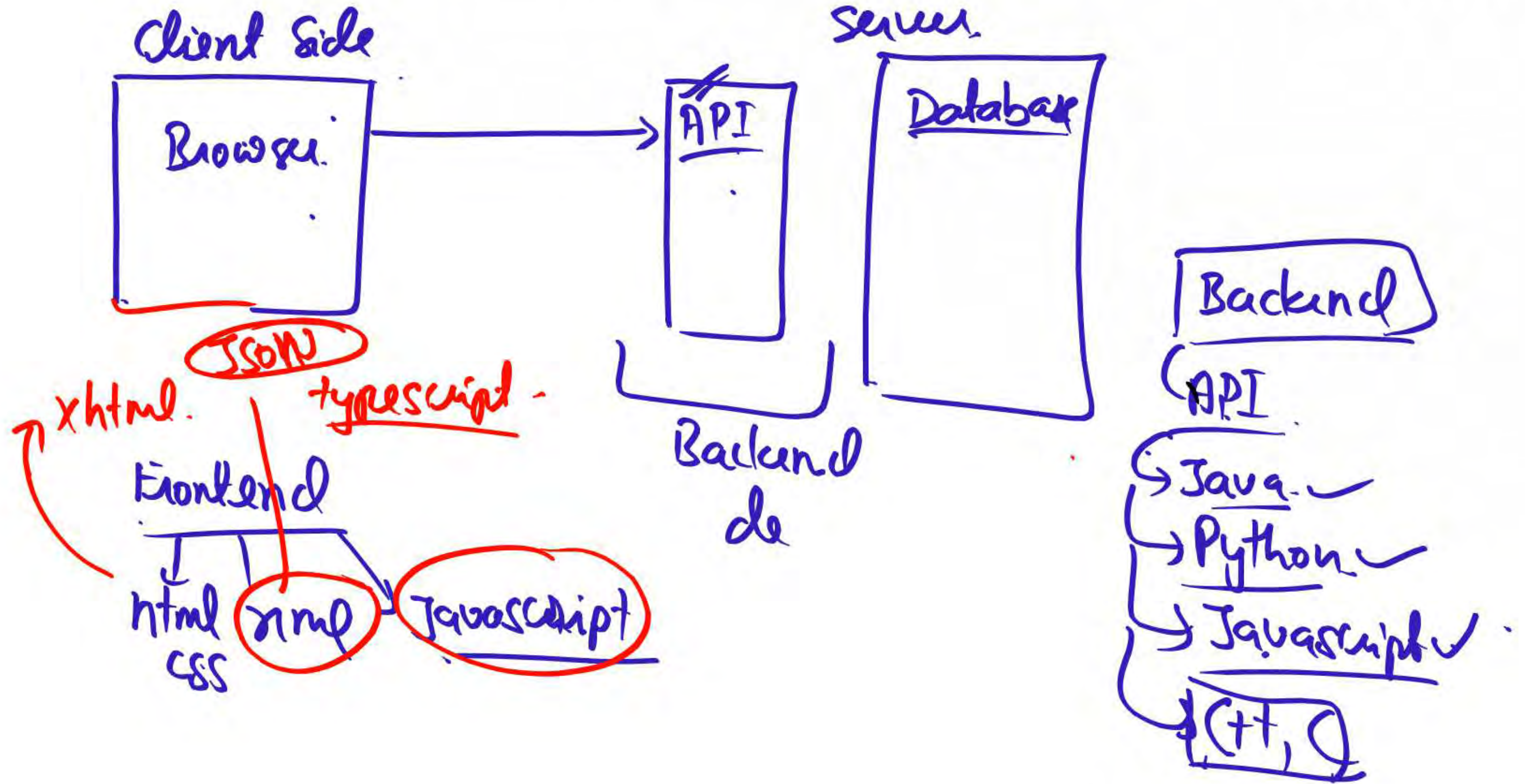
Application Software.

↓
ms office..

www $\Rightarrow$ world wide web
w3



web based application.



HTML

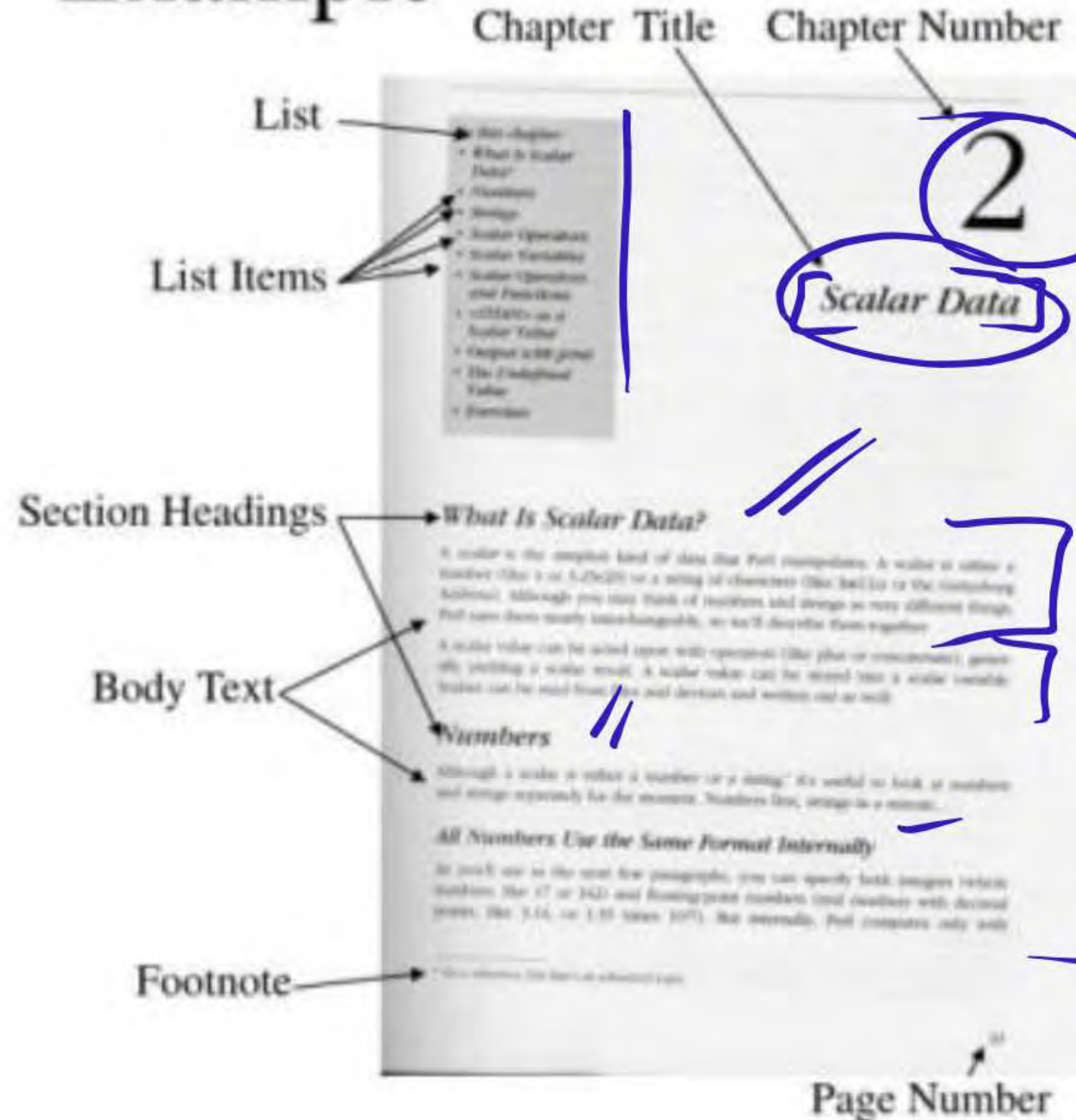
- What does HTML stand for?
 - HyperText Markup Language
 - HTML documents are simply text documents with a specific form.
- What does “markup” mean?
 - Comes from publishing.
 - Documents comprised of content and markup.
 - Content: actual information being conveyed.
 - Markup: side information about the content.

Markup

- Markup information is used to convey the role of different parts of the content and their relationships to each other.

Example

- Here is the first page of chapter two from “Learning Perl”.
- What are the roles of the different pieces of content?



HTML Tags

- Markup is indicated through the use of “tags”.
 - Tags are delimited by angle brackets `<>`
 - Beginning tags
 - Can also include attributes.
 - `<tagname a1=“v1” a2=“v2”>`
 - Ending tags
 - `</tagname>`
 - No attributes.

`<h>`

`</h>`

HTML Document Structure

- Entire document enclosed within <html> and </html> tags.

- Two subparts parts:

- Head

- Enclosed within <head> </head> tags.
 - Within the head further tags can be used to specify title of page, meta-information, etc.

- Body

- Enclosed within <body> </body> tags.
 - Within the body is the document content to be displayed.

<html>

<head> My Page. </head>

<body>

<p>

</body>

</p>

</html>

Xhtml.
[<!DOCTYPE> _____]

Basic Tag Types

- <title>
 - In the head of the document, provides a title for the page.
- <h1>, <h2>, <h3>
 - Different heading levels.
- <p>
 - Indicates paragraphs.
-

 - Indicates line breaks.

Tag Attributes

- Attributes specified in beginning tag modify and amend tag meaning.
- Each tag has a predefined set of meaningful attributes.
- Unrecognized attributes are ignored. → Error.
- Example:
 - Source file for image data.
 -

Anchors and Links

- Key feature of HTML is “linking” between documents.
- Portion of content can be specified as a link using the anchor tag <a>.
- The href attribute of the anchor tag provides the address of the linked document.
- Example:
 -

Anchors

- The target of a link can further specify a specific “anchor” to go to.
 - href=http://server/path/to/page#anchorname
- Defining anchors.
 - Name attribute of an <a> tag.
 -
 - ID attribution of any other tag.
 - <h1 id=“anchorname”>

Embedded Objects

- Other types of data can be embedded into an HTML document. } document
 - Most commonly: images
- Images indicated using the tag.
- The src attribute indicates where the image data can be found.
- Example:
 -

Function Not Format

- Markup the function of the content without specifying format of the content.
 - Allows end user to decide how function was mapped to appearance.
 - Example: `<em>` for emphasis.
- What are the advantages of doing this?
- What are the disadvantages?

Problems with HTML

- HTML became “polluted” with formatting specific tags and tag attributes.
 - Example: for bold.
- Different companies kept adding “custom” tags.
 - Why is this a problem?
- Although human-readable, not always easy for machine to parse and manipulate.
 - What could be hard for the machine but easy for the human?
 - Why might we want a more machine friendly document?

Computer Science.

XML to the rescue

JSON

IOT

↳ 1 class

<computer science>

<.IOT>

<class> 50 </class>

- Document format for structured data.
- Extensible markup languages
 - Domain-specific
 - Common agreements and standards.
- Separate form, meaning, and presentation.
 - Form defined by XML
 - Meaning defined by applications and XML-based standards.
 - Presentation defined separately from content.

Extensible Markup language
user defined tags

XML Document Structure

- XML Declaration `<?xml>`
 - Required
- Document Type Declaration `<!DOCTYPE>`
 - Optional
- Document Body
 - Required

XML Declaration

- <?xml version="1.0" encoding="UTF-8" ?>

- Must be first line of the document.
- The version attribute is required.
- Two optional attributes:

- encoding

- Standard name for character set encoding.
- UTF-8, UTF-16, US-ASCII, etc.

- standalone

- “yes” or “no”
- Indicates whether or not this XML document requires an external document type definition.
- Default is “no”
- Not required to be set.

web browser
(hello)

Server.

XML Declaration

- Valid declarations:

- `<?xml version="1.0" ?>`
- `<?xml version="1.0" encoding="UTF-16" ?>`
- `<?xml version="1.0" standalone="no" encoding="UTF-8" ?>`
- `<?xml version="1.0" encoding="US-ASCII" standalone="yes" ?>`

- Invalid declarations:

- `<xml version="1.0">`
- `<?xml version="1.0">`
- `<?XML Version="1.0"?>`
- `<?xml standalone="no" version="1.0"?>`
- `<?xml version=1.0" ?>`

DTD Declaration

- Two forms:

<!DOCTYPE *RootElement* SYSTEM *DTD_URI*?
[*InternalDeclarations*]? >

<!DOCTYPE *RootElement* PUBLIC *PUBLIC_ID URI*
[*InternalDeclarations*]? >

- Here “?” indicates optional.
- DTD specifies how the XML document can be validated.

Document Type Definition

Document Body

- The document body is comprised of markup and character data.
- Must contain at least one tag that encloses everything.
 - This is called the root element.

Markup vs. Character Data

- Markup is text in specific formats that defines the structure of the document and provides interpretation and parsing hints.
- Character data is everything else.

Markup

- XML declaration
- Document type declaration
- Element Tags
 - Start, End, and Empty
- Entity References
 - External, internal, and character.
- Comments
- CDATA
- Processing Instructions

XML Declaration and DTD

- Already reviewed what these markup elements look like.
- We'll get into DTD later.

Elements

- Elements are defined by a start and end tags.
 - Start tags: `<element_name attributes*>`
 - End tags: `</element_name>`
- Element Names:
 - Case sensitive
 - Can start with letter, underscore or colon
 - Can contain letters, underscore, colon, digits, hyphens, or periods.
- Everything between start and end tags is considered element content.

Attributes

- Form

attribute_name=“*value*”

attribute_name=‘*value*’

- Attribute names have same rules as element names and are also case sensitive.
- Values must be quoted
 - Can use either single- or double-quotes.

Reserved Attributes

- Some attributes are given special meaning and can be used with any element.
 - xml:lang
 - Provides language/character set information for everything within this element.
 - xml:space
 - Provides hint to parser whether whitespace is really significant or not.
 - Value can be “default” or “preserve”
 - Unfortunately, “preserve” is the default?!?!
 - xml:base
 - Controls base URI for document for resolving relative links and references.
 - xmlns
 - Used to define namespaces.

Empty Elements

- Sometimes elements are never intended to have content.
 - Think of HTML `<br>` tag.
 - If we wanted to make HTML XML-compliant:
 - `<br></br>`
 - Empty tag notation is shorthand for this:
 - `<element_name attributes*/>`
 - Example: `<br/>`

Comments

- Comments delimited by:
 - Start of comment: `<!--`
 - End of comment: `-->`
- Comment content ignored.
 - No meaning as markup or data.
- Restrictions
 - Cannot contain “--” except as part of delimiters.
 - Cannot be nested.
 - Cannot precede the XML declaration.
 - Cannot be within element tag (start, end, or empty)

Processing Instructions

- Also known as PI's
- Form:
 - `<?pi_name pi_data?>`
- Restrictions
 - No space between ? and *pi_name*
 - *pi_data* cannot contain “?>”
 - What *pi_data* means is defined externally.
- XML declaration is actually an example of a PI.
 - Some *pi_names* defined by specific standards.

Entities

- Given what we've seen so far, can you think of element content text that might cause problems?
 - What if the character data (i.e., element content) contains “<”?
 - Would be difficult to tell whether or not this was the start of a new tag or not.
 - Example:
`<equation> 3 < x + 2 </equation>`

Entities

- Entities provide a way to specify text which would be hard to specify directly.
 - “<” or “>” for example.
 - Unprintable characters.
 - Unicode characters for other languages.
- An XML parser interprets an entity reference and replaces it with its definition.
- Form:
&entity_name;



2 mins Summary



TEACHING
WALLAH

Topic

XML
XHTML
CSS } →

Java

3

THANK YOU